



Toxic vs. Non-Toxic

Plant Classification from Images

*A CNN-based binary image classifier to help
hikers, pet owners & parents stay safe outdoors*

The Problem: You Can't Tell Just by Looking

Accidental plant exposure is a recurring, preventable safety risk for hikers, pet owners, and parents.

- The look-alike problem is real — Poison Ivy & Virginia Creeper grow side-by-side
- Most people cannot identify toxic plants reliably in the field
- Exposure consequences range from severe rash to systemic poisoning



Stakeholders

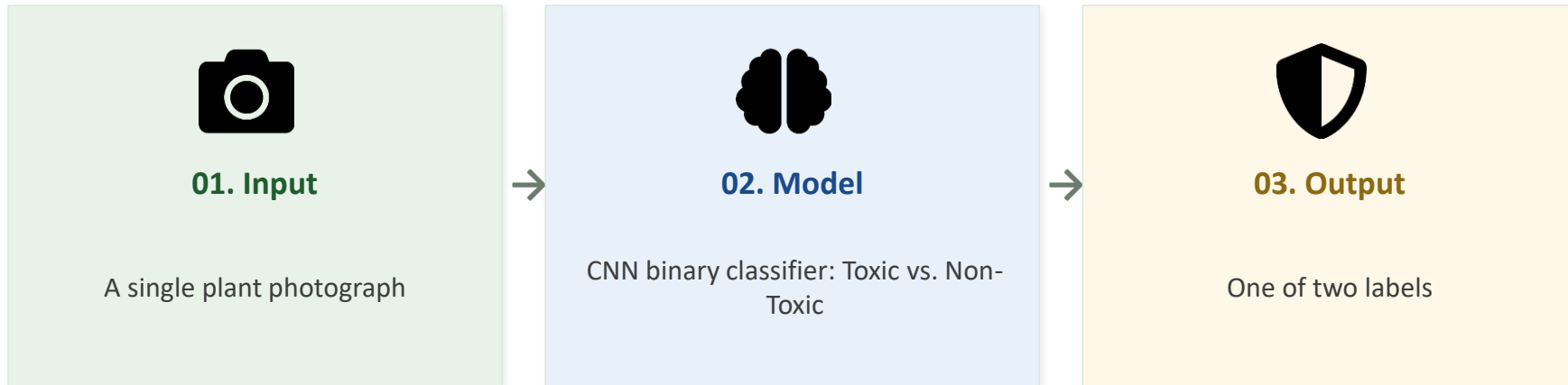
Hikers · Pet owners · Parents
Parks services · Agricultural educators



Success Metric

≥ 90% recall on the toxic class
A missed toxic plant is far costlier than a false alarm.

Binary Classification — Toxic or Not?



⚠ Key Constraints:

- With only 10 species, real world deployment will be limited.
- With only 1 machine, training the model is time consuming and memory intensive.

9,952 Plant Images — 10 Species

9,952

total images

10

plant species

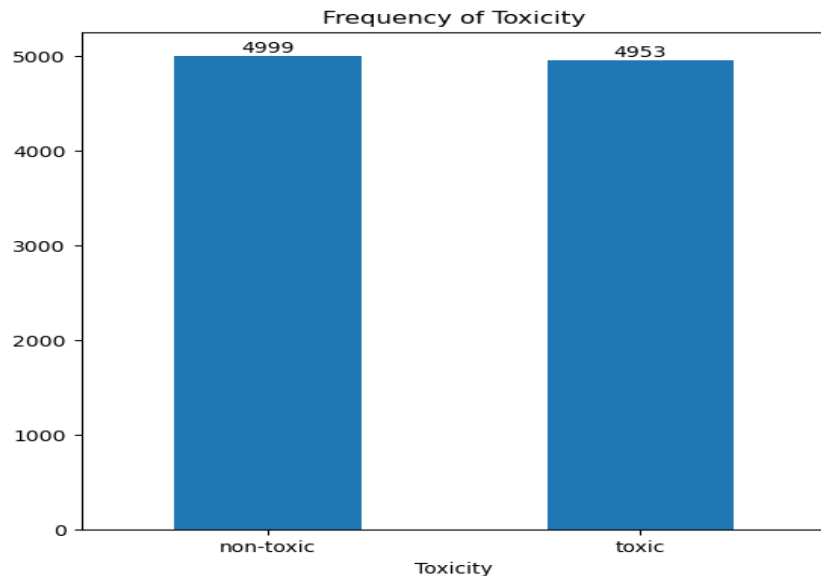
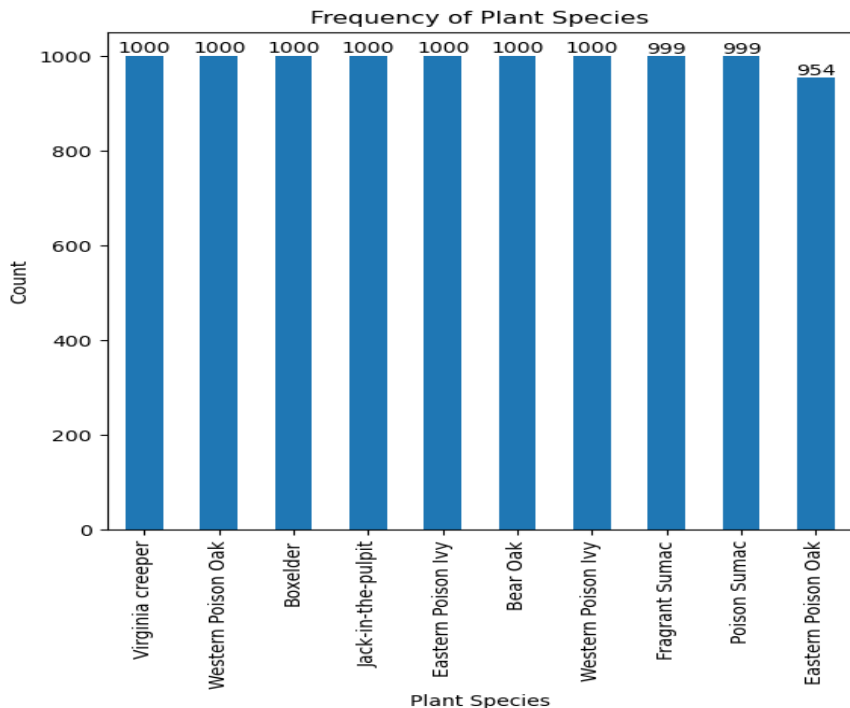
50/50 ~1,000

toxic vs non-toxic

images per species

Toxic Species	Non-Toxic Species
Eastern Poison Ivy	Virginia Creeper
Western Poison Ivy	Boxelder
Eastern Poison Oak	Jack-in-the-pulpit
Western Poison Oak	Bear Oak
Poison Sumac	Fragrant Sumac

A Well-Balanced Dataset



✓ 9 of 10 species have 999–1,000 images each. Overall split: 4,999 non-toxic / 4,953 toxic — no resampling required.

Cleaning Steps & Key Takeaway



01

Cleaning

No missing values in the data set. Unnecessary columns were removed from the dataset

02

Label Verification

Ground-truth toxicity labels verified against species assignments. No mis-labels found; Eastern Poison Oak slightly underrepresented at 954 images vs ~1,000 for others.

03

Adding features

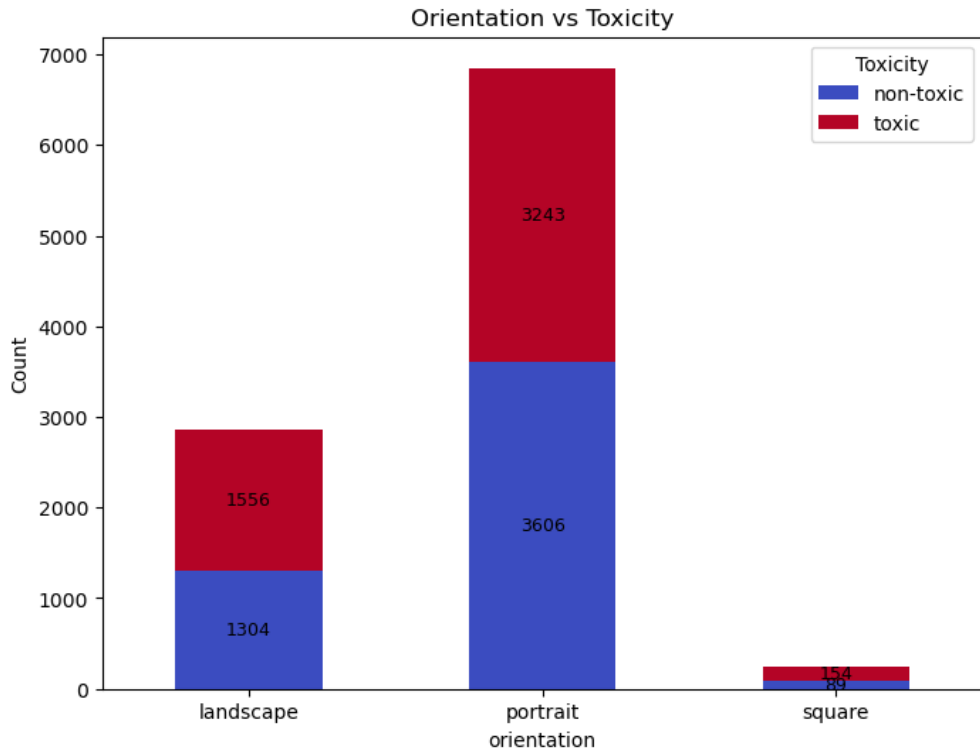
Added feature columns to metadata to allow for more Exploratory Data Analysis.

04

Augmentation Pipeline

Random crops, flips, rotation ($\pm 20^\circ$), color jitter, affine shear, perspective distortion & random erasing applied to training set only. Validation set: resize + center crop + normalize.

A Hidden Risk: Orientation Correlates with Toxicity



Data Leakage Risk

$$\chi^2 = 58.6, p < 0.001$$

Landscape photos are disproportionately toxic.
Portrait photos are disproportionately non-toxic.

The Fix

Heavy geometric augmentation (random flips, $\pm 20^\circ$ rotation, random resized crop) was added specifically to prevent the model learning orientation as a shortcut.

Without this fix, the model could reach high validation accuracy by classifying orientation — then fail completely on real-world photos taken at any angle.

Color Features: Minor Variation, No Leakage

Brightness

Similar distributions across both toxic and non-toxic classes. No exploitable signal.

Contrast

Consistent across species. No meaningful difference between toxic and non-toxic groups.

Hue

Broad overlap across all species — expected for green-dominant plant imagery.

Saturation ⚠

Poison Sumac trended slightly more saturated; Virginia Creeper & E. Poison Ivy trended lower. Difference too small to constitute leakage but noted for augmentation.

✓ Key takeaway: Color features present no systematic leakage risk. The hard work of classification must come from learned shape and texture features — not color shortcuts.

Why Transfer Learning & EfficientNet-B0?



1

Small dataset problem

~10k images is not enough to train a deep CNN from scratch. Transfer learning from ImageNet (1.2M images, 1k classes) gives us rich low-level visual features for free.

2

EfficientNet-B0 vs ResNet18

EfficientNet-B0 uses compound scaling to balance depth, width, and resolution — higher accuracy per parameter than ResNet18 on fine-grained visual tasks.

3

3-Phase Fine-Tuning

Freeze backbone → train head → unfreeze top 30% → full fine-tune. Prevents early layer destruction while allowing the final model to adapt texture & shape features to plant morphology.

Key Techniques

Class Weighting

1.3× on toxic class

Label Smoothing

$\epsilon = 0.05$

Mixup / CutMix

$\alpha = 0.4$, 25% of batches

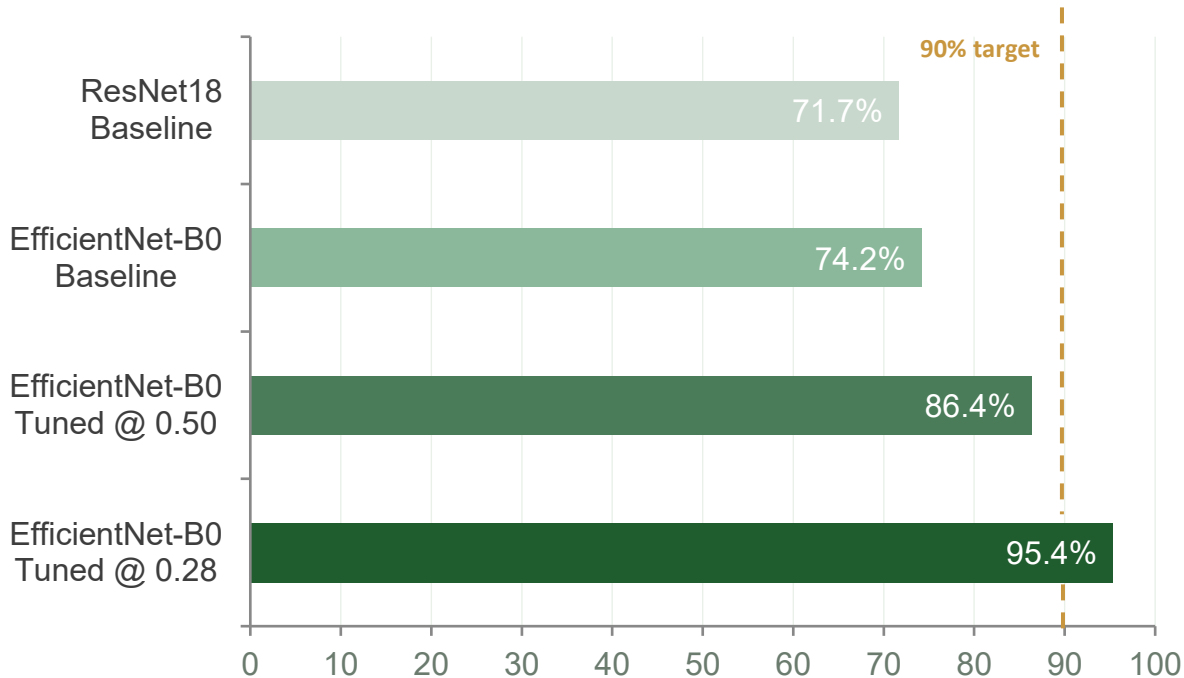
Temperature Scaling

Post-hoc calibration

Threshold Tuning

Maximize F2 score

Three Iterations — Each Building on the Last



ResNet18

Recall 71.7% · Acc 72.1%

Baseline

EffNet-B0 (frozen)

Recall 74.2% · Acc 75.4%

Baseline

EffNet-B0 (tuned, default threshold)

Recall 86.4% · Acc 83.7%

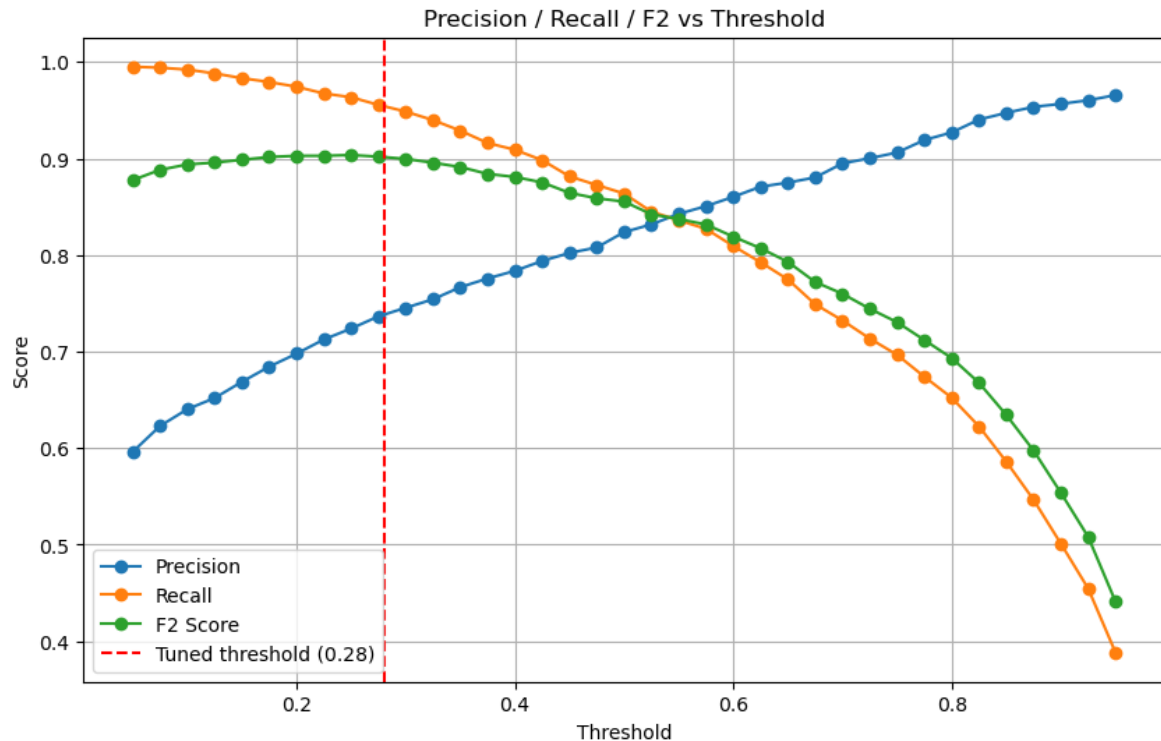
Better

EffNet-B0 + threshold

Recall 95.4% · Acc 80.3%

✓ **Target hit**

Lowering the Threshold to Prioritize Safety



How It Works

Default threshold (0.50) treats a missed toxic plant and a false alarm as equally costly — wrong for this use case.

We swept thresholds 0→1 and picked the value that maximized the F2 score (which weights recall 2× over precision).

Selected: threshold = 0.28

Recall (toxic) **95.4%**

Precision (toxic) **73.6%**

F2 Score **0.900**

Val. Accuracy **80.3%**

Validation Set Results at Threshold 0.28 (n = 1,991)

	Predicted: NON-TOXIC	Predicted: TOXIC
Actual: NON-TOXIC	633 (True Negative)	346 (False Positive)
Actual: TOXIC	47 (False Negative ⚠)	965 (True Positive)

95.4% Recall on toxic class

73.6% Precision on toxic class

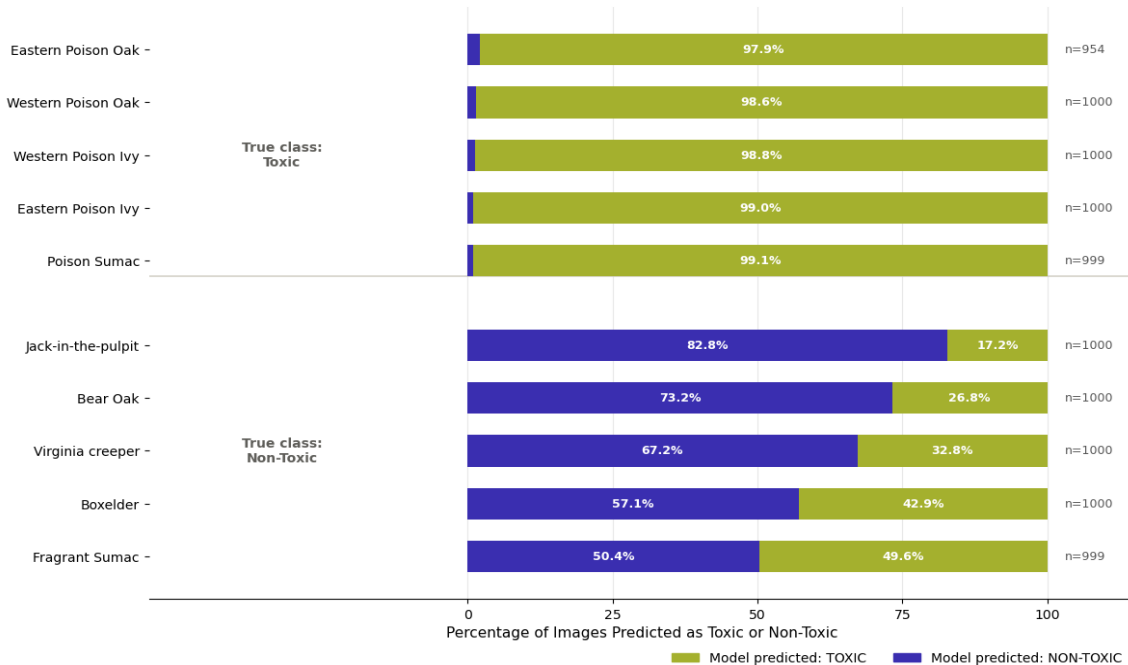
4.6% Miss rate (false negatives)

35.3% false alarm rate on non-toxic images (346 of 979) — intentional trade-off for 95.4% recall.

The 47 false negatives represent missed toxic plants — the error type we're most committed to minimising.

What the Binary Model Actually Does, Species by Species

**Toxic/Non-Toxic Binary Classifier Predictions,
Broken Down by Plant Species**



Toxic species: 98–99% correctly flagged TOXIC



Non-toxic species: Fragrant Sumac 50%, Boxelder 57% false alarms

Three Recommendations for the Client

01

Deploy with an honest caveat

Launch the tuned model (threshold 0.28) for its intended safety use case — but pair every 'toxic' prediction with a brief note that a “toxic” result means “please treat this with caution,” not “this is definitely toxic.”

02

Build the species follow-up model next

Since toxic-class recall is already excellent (98–99%), the highest-value next investment is a second-stage species-level model to reduce false alarms on non-toxic look-alikes — not further binary threshold tuning.

03

Position as a first-pass filter, not an authority

Market the tool as a quick screening aid ('treat this plant with caution') while always directing users to professional resources — extension offices, poison control — for any plant to be handled or ingested.

What Comes Next



More data for weak species

Targeted collection/augmentation for Fragrant Sumac, Boxelder & Virginia Creeper — the three non-toxic species with highest false-alarm rates.



Species-level follow-up model

A second-stage classifier that identifies which of the 10 species an image most resembles, enabling users to sanity-check a 'toxic' flag.



Field photo validation

Validate performance on real-world user-submitted photos — messier, more varied lighting & backgrounds than the curated dataset.



Expand species coverage

Beyond the current 10 species to cover more regionally common toxic plants — the architecture supports it without a full redesign.



Add geolocation & seasonal features

Augment the species level image model with metadata — location and time of year



Thank You

Questions & Discussion